

Tutorial 1 (Experimental design and CRD/One-way ANOVA)

Question 1

Experiment Description

Three groups of students, 5 in each group, were receiving therapy for severe test anxiety in an experiment design as a Completely Randomised Design. Group 1 received 5 hours of therapy, group 2 received 10 hours, and group 3 received 15 hours. At the end of therapy, each subject completed an evaluation of test anxiety (the dependent variable in the study). Did the amount of therapy have an effect on the level of test anxiety?

The three groups of students received the scores on the Test Anxiety Index (TAI) at the end of treatment shown in the table below and in the csv file called “test_anxiety.csv”. You will need to use R at some point.

Group 1	Group 2	Group 3
48	55	51
50	52	52
53	53	50
52	55	53
50	53	50

1. Identify the following based on the experiment description:
 - a. Treatment factor
 - b. Factor levels
 - c. Treatments
 - d. Response
 - e. Experimental units
 - f. Number of replicates
2. What is the treatment structure and how has randomization been conducted?
3. What would be the problem if we said group was the experimental units?
4. Write down the model equation used for analysing data from this type of experiment. Be specific.
5. Calculate the following by hand and note down the R code you could use to calculate the first three:
 - a. Overall mean
 - b. Treatment means
 - c. Treatment effects
 - d. Estimated error variance
6. Write out the separate model equations for each treatment.
7. What is the error term for observation Y_{34} ?
8. What is the standard error of treatment means? Of differences between treatment means? Calculate without using R (you won't have R in the exam).

9. What are the assumptions of this model and how would you check the validity of each of these? Remember we can check assumptions before and after model fitting.
10. Plot the data per treatment as box plots with the points overlayed. Select the options that apply:
 - a. The boxplots indicate that there are severe outliers.
 - b. It is hard to tell if the data are normally distributed because the sample sizes are so small.
 - c. The heights of the boxplots look comparable.
11. Besides boxplots, what other type of visualization can be used to check the assumption of normality?
12. Which assumption(s) do we need to check after model fitting as well? Why is that?
13. What is the ratio of largest to smallest standard deviations of the treatments?
14. If the sample standard deviations are different that means that the groups do not have equal population variance. True or false? Motivate your answer.

Question 2

A food scientist is investigating whether four different baking temperatures affect the average rise height (cm) of a particular type of bread.

- The four temperatures are: 180°C, 190°C, 200°C, and 210°C.
 - For each temperature, the scientist prepares 12 loaves of bread, using the same recipe and ingredients.
 - Each loaf is baked independently in the oven at the assigned temperature.
 - After baking, the rise height of each loaf is measured once.
 - The loaves are baked in a random order.
1. How many replicates does this experiment have?
 2. What is the experimental unit?
 3. Why is it important that each loaf is baked independently?
 4. TRUE OR FALSE: Randomising the baking order helps ensure that differences in rise height are due to temperature rather than time-related effects.
 5. Is this experiment balanced or unbalanced?
 6. Suppose the scientist measured the rise height of each loaf three times and recorded the average. Would this change the number of replicates? Explain.
 7. If the loaves had been baked one temperature per day (all 180°C on Day 1, all 190°C on Day 2, etc.), what potential problem could arise?
 8. State the null and alternative hypothesis for analysing this experiment using ANOVA.
 9. Write down the model equation used for analysing data from this type of experiment. Be specific.
 10. Use the following code in R to simulate data and check the assumptions.

```
set.seed(123) #this is very important, it ensure you get the same data every time
```

```
# Factor: baking temperature
```

```
# Creates a factor representing the four baking temperatures, with 12 loaves  
(replicates) per temperature.
```

```
temp <- factor(rep(c("180", "190", "200", "210"), each = 12))
```

```
# Simulated rise heights (cm)
```

```
# each rnorm line generate values from a normal distribution with the specified  
mean and sd
```

```
rise <- c(  
  rnorm(12, mean = 6.0, sd = 0.5),  
  rnorm(12, mean = 6.4, sd = 0.5),  
  rnorm(12, mean = 6.3, sd = 0.5),  
  rnorm(12, mean = 5.8, sd = 0.5)  
)
```

```
# put together in a dataframe
```

```
bread <- data.frame(temp, rise)
```

More questions

To answer these questions, use the lecture slides and notebook.

2. Explain the difference between an observational study and an experiment.
3. What is one key principle of experiments that allows the establishment of causal relationships?
4. Why is it difficult to establish causal effects through observational studies?
5. Define the following terms in terms of experimental design:
 - a. Response
 - b. Factor
 - c. Treatments
 - d. Experimental units
6. What is an experiment called where treatments are combinations of factor levels?
7. Why is having more than one response value per experimental unit problematic?
8. What design principle can be used when experimental units are not homogenous? How is randomization applied in this procedure?

9. Suppose a researcher conducts 10 independent hypothesis tests, each at a 5% significance level. If all null hypotheses are true, calculate the probability that the researcher makes at least one Type 1 error.